

Observed changes in southwest monsoon rainfall over India during 1901–2011

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ABSTRACT: The trends and epochal variability of southwest monsoon over the country as a whole and four homogeneous regions are examined using monthly rainfall data (1901–2011) of 640 political districts of India. The district rainfall data is computed from station rainfall data. The same station data is used to analyse the trends in the frequency of rainfall events of different intensities for examining extreme rainfall events. The existence of the multidecadal epochal variability of rainfall is clearly established in the all-India monsoon rainfall as well as monsoon rainfall over the four homogenous regions. However, over different homogenous regions, the phases of multidecadal variability are found to be different. Principal component analysis brings out Northeast India (NEI) rainfall as more dominant mode for all-India rainfall. Significant decrease in southwest monsoon rainfall over NEI is observed during the post 1950 period. Decreasing trends are also observed over the monsoon core region during the post-1950 period. Over these regions, monsoon rainfall has increased significantly during the pre-1950 period. It has been shown that the decreasing trend in monsoon rainfall during the post 1950 period is the result of multidecadal epochal variability. Geographical regions that experienced significant changes in the frequency of days of rainfall with different intensities are also identified. Significant change/turning points are also detected in the southwest monsoon rainfall. Frequency of moderate rainfall events ($5 \text{ mm} \leq \text{daily rainfall} < 100 \text{ mm}$) decreased significantly during the period 1951–2010 over the monsoon core region of India whereas no significant changes are observed in the frequencies of heavy (daily rainfall $> 100 \text{ mm}$) or very heavy rain (daily rainfall $> 150 \text{ mm}$) during the southwest monsoon season. Climatic shift or change point in monsoon rainfall in India is also detected by an established statistical test.

KEY WORDS monsoon rainfall; trends; extremes; climate change; change point

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1. Introduction

The climate of South Asia is largely dominated by the Asian monsoon circulation. Most of the annual rainfall over India is received during the 4 months of southwest monsoon season (June–September). Though nearly 75% of annual rainfall of the country is received during the southwest monsoon season, there is a large spatial variability in the rainfall distribution. Some regions are highly dominated by the southwest monsoon season. For example, most of the annual rainfall over Northwest India occurs during the southwest monsoon season. Northeast India (NEI) region receives good amount of rainfall during the pre-monsoon and summer monsoon seasons. However, southeast Peninsular India receives significant amount of rainfall during the post-monsoon season in addition to summer monsoon season. The northern parts of western coast receive about 90% of annual rainfall during the southwest monsoon season with Gujarat region receiving maximum rainfall (95%). Southwest monsoon rainfall over India exhibits variability in all

time scales from diurnal to multidecadal. In some years, the year to year variability of monsoon rainfall leads to floods and droughts over the country. Thus, a detailed analysis of changing monsoon rainfall pattern and also its intensity using updated and reliable data is urgently needed.

The year to year variability of monsoon rainfall has been studied by many researchers during the past six decades (Pramanik and Jagannathan, 1954; Parthasarathy and Dhar, 1978; Parthasarathy, 1984; Moolley and Parthasarathy, 1984; Rupa Kumar *et al.*, 1992; Parthasarathy *et al.*, 1993; Pant and Rupa Kumar, 1997; Kripalani and Kulkarni, 2001 and others). As rainfall has high temporal variability than other climate parameters like temperatures, 30 years of data are at least required for preparing climate normal of rainfall. Even though the World Meteorological Organization (WMO, 2008) prescribes minimum 30 years of data for rainfall normal, this criterion is not adequate for the monsoon rainfall as decadal rainfall variability is a significant factor. Guhathakurta and Rajeevan (2008) have created a homogeneous rainfall data series using 1476 stations of India for the period 1901–2003. Using the same data, they have shown that the all-India rainfall has no long-term

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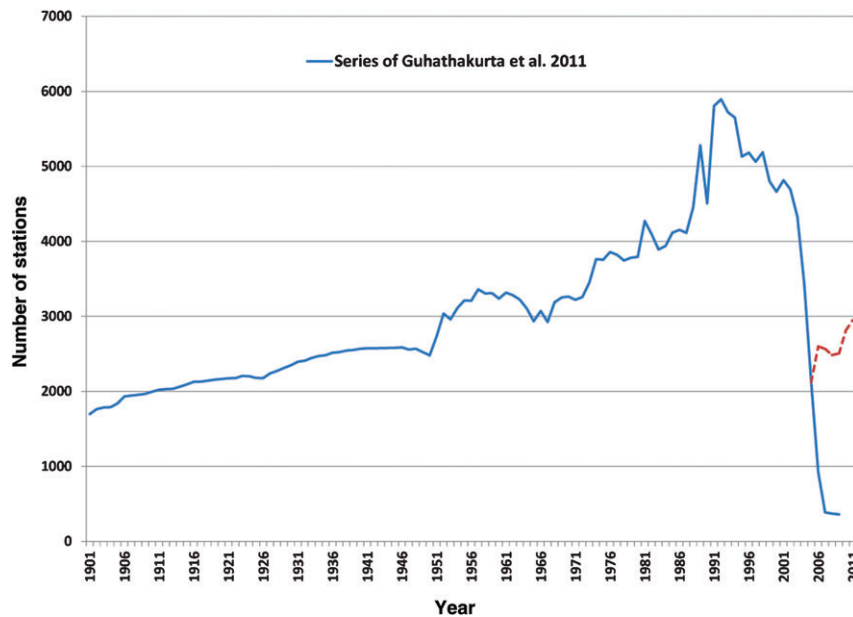


Figure 1. Yearly frequency of the number of stations used for analysis. The frequency of stations from 1901 to 2004 shown in solid line and from 2005 to 2011 shown in dotted line are considered. The series shown in solid line from 2005 to 2011 is not considered as the frequency is very poor in those years.

trend for the monsoon rainfall or for any monsoon months individually. However, in their analysis the network of stations could represent only 458 districts of more than 640 political districts of India. The existence of 30-year epoch, which was earlier mentioned by Parthasarathy *et al.* (1993) and Rupa Kumar *et al.* (1992), of all-India rainfall was indicated in their study. Owing to the presence of high spatial variability of Indian monsoon rainfall, these 30-year epoch (1901–1930, 1931–1960, etc.) were not in the same phases for the four broad homogeneous regions of India. The recent studies have suggested the weakening of monsoon circulation over India (Joseph and Simon, 2005; Chung and Ramanathan, 2006; Ramesh and Goswami, 2007; Bawiskar, 2009; Naidu *et al.*, 2011). These studies are based on the dataset (1951–2004/2006). Southwest monsoon rainfall is known to exhibit significant multidecadal variations (Joseph, 1976; Pant and Rupa Kumar, 1997; Guhathakurta and Rajeevan, 2008). The analysis of rainfall data for the period 1951–2004 shows decreasing trend, as the decades 1931–1960 were wet decades and the decades 1961–1990 were followed with below normal rainfall. Thus, when the rainfall data of 1951–2004 is examined, decreasing trend is obvious. Long-term changes of global land monsoon precipitation were also assessed by Zhang and Zhou (2011) using three sets of rain gauge precipitation data for the period 1901–2001. Their study revealed that precipitation averaged over Northern Hemisphere has decreased during the period and rainfall for land monsoon over different regions of the world has a decreasing trend. There were also reporting of overall increasing trend from 1901 to 1955 and then a decline trend up to the present time.

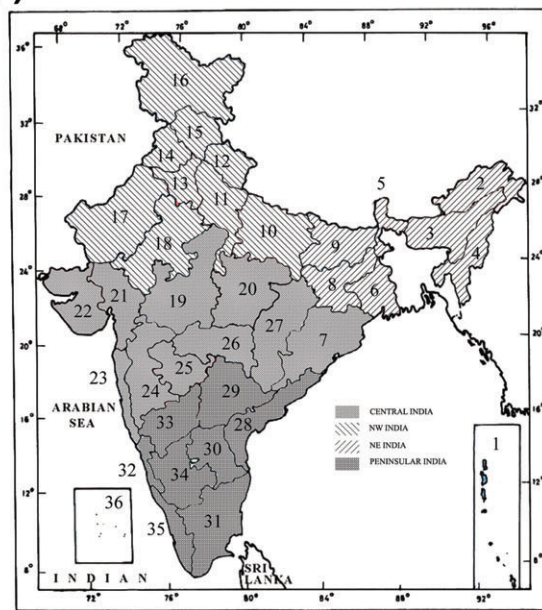
In this study, we have used new rainfall data series (1901–2011) of India Meteorological Department (IMD)

to represent rainfall variations in more than 640 political districts of India. Trends and epochal variability of south-west monsoon rainfall for each of the four homogeneous regions and also all-India were analysed in detail. Use of rainfall data from all the 640 districts ensures spatial coverage. Trend analysis of rainfall is examined for both monotonic trends as well as non-monotonic or change point detection. For the first time, climatic shift or change point in monsoon rainfall in India is also detected by an established statistical test. The analysis of frequency of rainfall in different categories based on the intensity is very much needed for agriculture and water resource management. A detailed analysis of observed changes in the frequency of rainfall days (in three categories, *viz.* wet days, rainy days and heavy rainfall days) for all the stations under each small geographical region (36 meteorological subdivisions as shown in Figure 2(a)) was also carried out. The significant changes in the frequencies of rainfall days of different intensities of stations in each meteorological subdivision between the period 1901–1950 and 1951–2011 and during the complete period 1901–2011 were also worked out to identify the region of significant changes.

2. Data and methodology

Recently, Guhathakurta *et al.* (2011a) have generated new rainfall series for the districts, meteorological subdivisions and all-India using quality controlled rain gauge data. However, there is sharp decline in the availability of stations from 2006 onwards. We have taken the monthly rainfall data from 1901 to 2004 from this time series and the data from 2005 to 2011 are taken from the District Rainfall Monitoring Scheme (DRMS) data available at the Hydromet Division of the India Meteorological Department (IMD), Pune.

(a)



- | | | |
|-----------------------------|--------------------------|------------------------------|
| 1. Andaman & Nicobar Island | 13. Har Delhi Chandigarh | 25. Marathwada |
| 2. Arunachal Pradesh | 14. Punjab | 26. Vidarbha |
| 3. Assam & Meghalaya | 15. Himachal Pradesh | 27. Chhatisgarh |
| 4. N M M T | 16. Jammu & Kashmir | 28. Coastal A P |
| 5. S H W B & Sikkim | 17. West Rajasthan | 29. Telengana |
| 6. Gangetic W. Bengal | 18. East Rajasthan | 30. Rayalseema |
| 7. Orissa | 19. West M P | 31. Tamil Nadu |
| 8. Jharkhand | 20. East M P | 32. Coastal Karnataka |
| 9. Bihar | 21. Gujarat Region | 33. North Interior Karnataka |
| 10. East U P | 22. Saurashtra & Kutch | 34. South Interior Karnataka |
| 11. West U P | 23. Konkan & Goa | 35. Kerala |
| 12. Uttaranchal | 24. Madhya Maharashtra | 36. Lakshdweep |

(b)

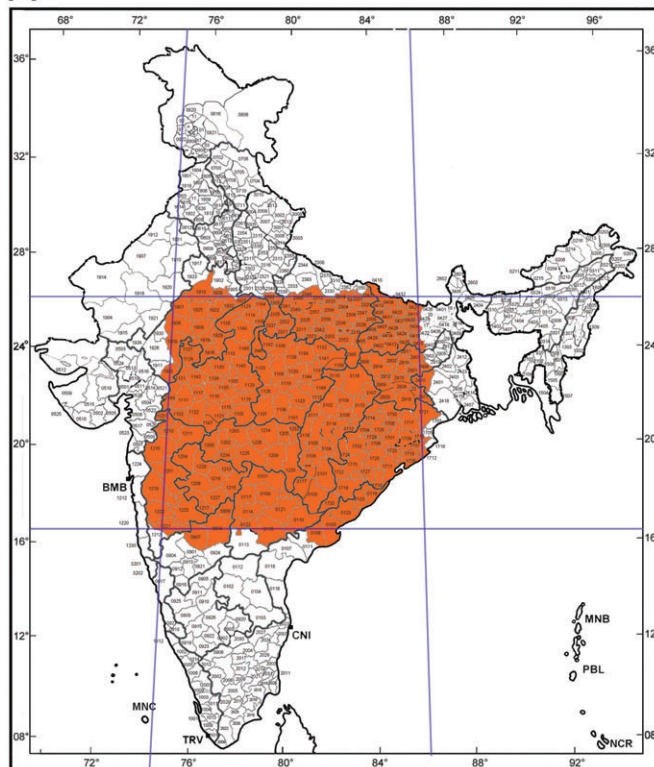


Figure 2. (a) The four homogeneous regions, viz. Northwest India, Central India, Northeast India and peninsular India considered by India Meteorological Department. (b) MCRI containing 230 districts within 74.5° – 86.5° E and 16.5° – 26.5° N.

Figure 1 shows the number of stations used for the present analysis. The solid line shows the number of stations in the original series of Guhathakurta *et al.* (2011a). There is a sharp fall from the year 2005 onwards in the number of stations (solid line) and thus the data from 2005 onwards of this series is not considered here. From 2005 onwards, we have used the DRMS stations, which are shown as dotted line. The methodology used for computation of district (small geographical region) rainfall is the simple arithmetic mean of station values whereas for subdivision (having larger geographical region than district) rainfall it is the district area-weighted mean. The average number of stations used for this period is around 3000. As it has been already established the existence of 30-year epoch in Indian rainfall (Joseph, 1976; Pant and Rupa Kumar, 1997; Guhathakurta and Rajeevan, 2008, etc.), the present analysis considers the rainfall received over India in following six epochs: (1) 1901–1930, (2) 1931–1960, (3) 1961–1990, (4) 1991–2011, (5) 1901–1950, (6) 1951–2011 and also (7) the complete period 1901–2011.

For the present analysis, rainfall series for the following three cases were computed and analysed: (1) the all-India area-weighted rainfall (calculated from area-weighted subdivision rainfall). (2) Rainfall for the four homogeneous regions (calculated from area-weighted subdivision rainfall) *viz.* Northwest India, Central India, Northeast India and Peninsular India considered by IMD (Figure 2(a)). (3) Monsoon Core Region of India (MCRI) containing 230 districts within 74.5°–86.5°E and 16.5°–26.5°N (Figure 2(b)). The districts (smallest geographical areas) within this region are shaded. Rainfall for this MCRI is calculated from the area-weighted rainfall of all the 230 districts. The purpose of selecting the MCRI is to examine the results of trend analysis of mean southwest monsoon rainfall. Also, the frequency of rainfall events for the period 1951–2000 using station data was compared with that obtained by Goswami *et al.* (2006) using gridded rainfall data for the same period and for the same region. Our analysis is limited to southwest monsoon season of India which includes four months, *viz.* June, July, August and September.

In addition, for the analysis of frequency of days of rainfall with different intensities, we have also considered 36 meteorological subdivisions which help to identify the regions showing significant changes for administrative uses in management rather than contour analysis. Rainfall intensities are categorized as (1) wet day if rainfall of the day is greater or equal to 0.1 mm, (2) rainy day if rainfall is more than 2.4 mm but less than 6.5 cm (this category gives the rainfall of moderate intensity) and (3) heavy rainfall day when rainfall is 6.5 cm or more. These categories are almost similar to that used by IMD. Here our category of heavy rainfall includes very heavy and extremely heavy also. From the daily rainfall of each station, the frequency of days of rainfall of these intensities are calculated. The least square linear fit is used to examine the existence of monotonic trend in the rainfall time series data and the statistic used for testing the significance is the Student's

t-distribution. For the analysis of trends in the frequencies of different rainfall events, we have used the well-known Mann–Kendal nonparametric trend test.

Climate and human activities have strong influence on hydrological processes. Some particular climatic phenomena as well as some human activities may cause sudden changes in hydrological processes that lead to abrupt change in the hydrological time series. Thus, studying whether there are any abrupt changes in the time series or not, and identifying the locations of change points, if any, are both essential for checking the stationarity and consistency of rainfall time series. Non-monotonic or abrupt change or change point (Climate jump) is the sudden change in climate pattern, also known as Climate jump. We have used Pettitt test to identify a change point in a time series, that assumes that the observations form an ordered sequence Pettitt (1979).

The test used here for change-point detection developed by Pettitt (1979) is a nonparametric test, which is useful for evaluating the occurrence of abrupt changes in climatic records (Tarhule and Woo, 1998; Smadi and Zghoul, 2006). One of the reasons for using this test is that it is more sensitive to break in the middle of the time series (Wijngaard *et al.*, 2003). The statistic used for the Pettitt's test has been explained by Dhorde and Zarenistanak (2013), Kang and Yusof (2012) and many others.

It is computed as follows:

The two samples $(x_1, x_2, \dots, x_t)$ and $(x_{t+1}, x_{t+2}, \dots, x_T)$ come from the same population. The test statistic $U_{t,T}$ is given by

$$U_{t,T} = \sum_{i=1}^t \sum_{j=t+1}^T \text{Sgn}(x_i - x_j) \quad (1)$$

The most significant change point is found where the value $|U_{t,T}|$ is max: $K_T = \max |U_{t,T}|$ and the significant level associated with K_T^+ or K_T^- is determined approximately by:

$$\rho = \exp\left(\frac{-6K_T^2}{T^3 + T^2}\right) \quad (2)$$

If ρ is smaller than the specific significance level, e.g. 0.1 in this study, the null hypothesis is rejected. In other words, if a significant change point exists, and the time series is divided into two parts at the location of the change point.

There are many other statistical tests like Leith's (1978) test, 'Y' test (Yamamoto *et al.*, 1985) and Table-look-up test (Guhathakurta and Srivastava, 2004) for change point detection.

3. Variability and trends of all-India rainfall

3.1. Decadal variability

The all-India monthly, seasonal and annual rainfall series were constructed based on the area-weighted rainfall of all the 36 meteorological subdivisions of the country. The present rainfall data series (abbreviated as IMD DIST DATA) is compared with three other data series of the IMD, *viz.* IMD operational rainfall data (used for real-time

Table 1. Comparison of mean and standard deviation (SD) of all-India southwest monsoon rainfall values based on present rainfall data (IMD DIST DATA) with various IMD rainfall dataset for the common period 1951–2000.

Various rainfall dataset of IMD	Mean (mm)	SD (mm)
IMD DIST DATA	888.26	89.00
IMD OP DATA	897.09	93.68
IMD 1 × 1 DEG GRID DATA	936.13	87.68
IMD 0.25 × 0.25 DEG GRID DATA	851.34	81.80

monitoring by IMD, abbreviated as IMD OP DATA), the 1 × 1 degree IMD daily gridded rainfall dataset (Rajeevan *et al.*, 2006) (abbreviated as IMD 1 × 1 DEG GRID DATA), and the new 0.25 × 0.25 degree gridded dataset (Pai *et al.*, 2013) (abbreviated as IMD 0.25 × 0.25 DEG GRID DATA). All the data series can be better understood by comparing the means and standard deviations of the southwest monsoon rainfall (June–September) for the uniform period 1951–2000 (Table 1). Both the rainfall data series IMD DIST DATA and IMD OP DATA which are generated from the station data give nearly similar values in both mean and standard deviation. However, the mean of IMD 1 × 1 GRID DATA is about 40 mm more and the mean of IMD 0.25 × 0.25 GRID DATA is around 40 mm less compared to the IMD OP DATA. For the IMD 0.25 × 0.25 GRID DATA, standard deviation is much lesser than the other datasets. Figure 3 shows the all-India southwest monsoon rainfall (in mm) for four datasets for the period 1951–2000. Large variation between the two gridded

dataset are seen in many years. But the climatology and the time evolution of both the station data series are consistent.

In order to study the decadal variability of all-India southwest monsoon season rainfall, decadal mean of all-India monsoon rainfall is calculated. Figure 4 shows the decadal mean of all-India monsoon rainfall shown as percentage departure from all-India mean southwest monsoon rainfall. The decadal values are compared with the other three monsoon rainfall data series, *viz.* IMD OP DATA, IMD 1 × 1 GRID DATA and IMD 0.25 × 0.25 GRID DATA. The long-term mean as well as decadal means of the present data series, *i.e.* IMD DIST DATA, agree well with the IMD OP DATA. But the long-term mean of IMD 0.25 × 0.25 GRID DATA is much lower and also the mean values for the decades 1921–1930 and 1961–1970 are opposites in sign to that of other three datasets. Multidecadal variability of monsoon rainfall is clearly noticed with the first two decades, *i.e.* 1901–1910 and 1911–1920, showing strong negative departures. The next five decades, *i.e.* 1921–1930, 1931–1940, 1941–1950 and 1951–1960, were in the positive phase (rising phase) whereas all the last four decades, *i.e.* 1971–1980, 1981–1990, 1991–2000 and 2001–2010, were in the negative phase (decreasing phase). The analysis clearly shows that the negative phase of all-India monsoon rainfall in the recent decades could be part of the natural long period variability and can be explained by periodicity of the all-India rainfall time series. We have performed a polynomial fitting (of order 3) to these decadal means. Figure 4 also shows the fitted polynomial

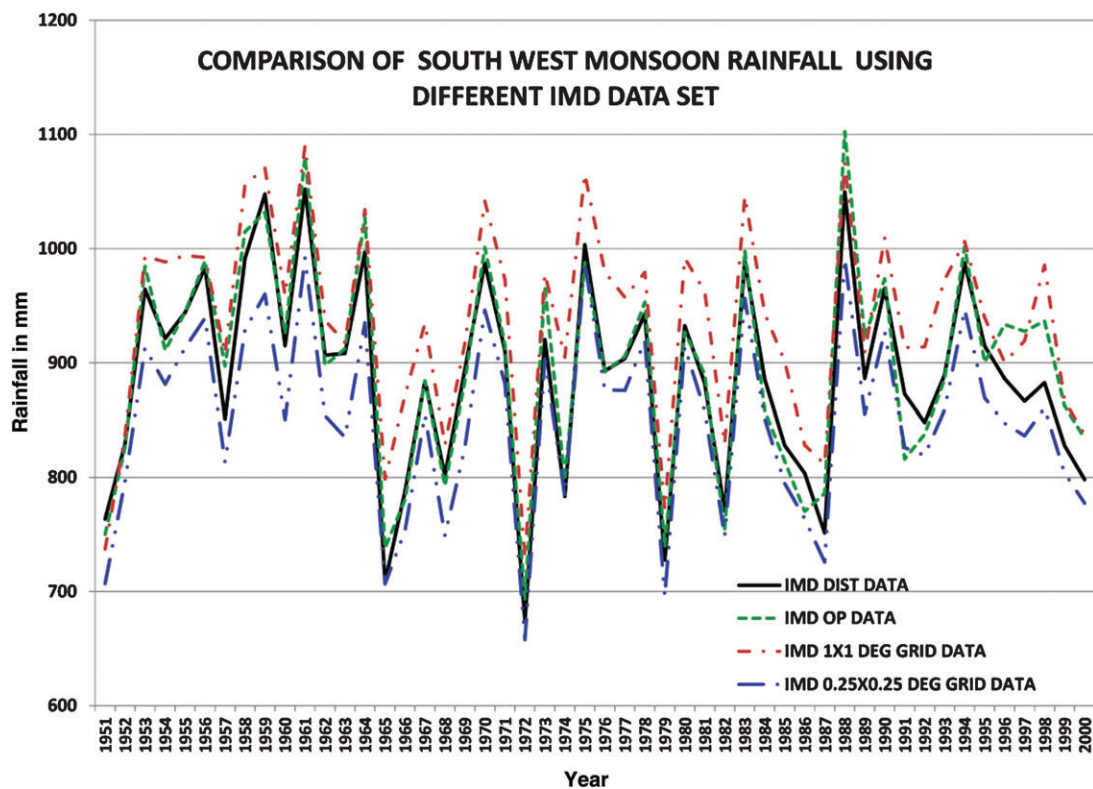


Figure 3. All-India southwest monsoon rainfall during 1951–2000 for four IMD datasets.

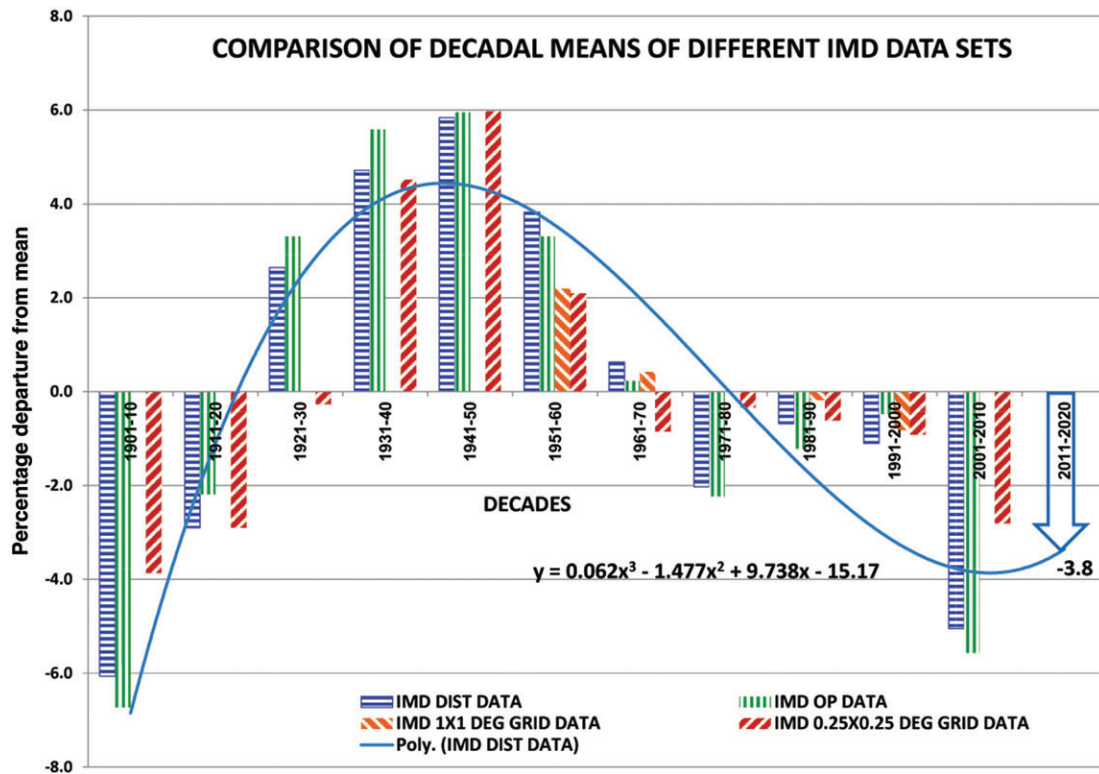


Figure 4. Decadal variation of all-India monsoon rainfall (IMD DIST DATA) along with polynomial fit line. Percentage departure values are obtained from the mean monsoon rainfall value of 887.1 mm calculated for the period 1901–2011. Decadal variations of all-India monsoon rainfall based on other three datasets are also shown.

Table 2. Trends (mm/season) in southwest monsoon rainfall during the 30-year epoch for four homogeneous regions and all-India.

	Trends in 30-year epoch				Trends in 50 years		Trends in 111 years
	1901–1930	1931–1960	1961–1990	1991–2011	1901–1950	1951–2011	1901–2011
All India	4.03*	0.23	−0.32	−1.82	2.79*	−1.00	−0.09
Northwest India	0.90	3.73*	−0.29	−0.72	0.92	−1.01	0.03
Central India	2.80	−0.30	−1.81	2.20	3.25*	−0.62	0.12
Northeast India	14.78*	−7.62*	2.67	−15.28*	6.88*	−3.05*	−1.99*
Peninsular India	−1.46	3.06	−0.25	0.77	−0.22	−0.32	0.39

Asterisk indicates significant trends (95%).

line along with the equation of the polynomial. Based on the polynomial fitting, the decade 2011–2020 is also likely to be in the negative phase with decadal mean percentage departure value as around −3.8%. The decade 1901–1910 was the decade of lowest rainfall (departure of −6% from long-term average) whereas the decade 1951–1960 was that of highest rainfall (departure of +6% from long-term average). The most recent decade 2001–2010 has also below normal rainfall (departure of −5.1% from long-term average). Thus, significant rising trend during 1901–1950 and decreasing trend during 1961–2010 are observed. Earlier studies by Pant and Rupa Kumar (1997) and Guhathakurta and Rajeevan (2008) have also found similar results of 30 years of alternating sequences of dry and wet periods. Our results extend up to the period 2011 and are based on a reliable network of more number of rain gauge stations.

3.2. Monotonic trends

Monotonic trends are generally gradual changes that are either increasing or decreasing with no reversal of direction.

3.2.1. Epochal trends

The southwest monsoon rainfall trends for all-India and four homogeneous regions (Figure 2(a)) for the epochs 1901–1930, 1931–1960, 1961–1990 and 1991–2011 were performed and the results are summarized in Table 2. The all-India rainfall has shown increasing trend in 1901–1930 and 1931–1960 and decreasing trend in 1961–1990 and 1991–2011. It is noted that while significant trends are observed during different epochs for all-India, Northwest India, Central India and Northeast India, the trends for Peninsular India have no statistical significance. Hence Peninsular India is the most stable monsoon region of all the regions. It clearly shows that four

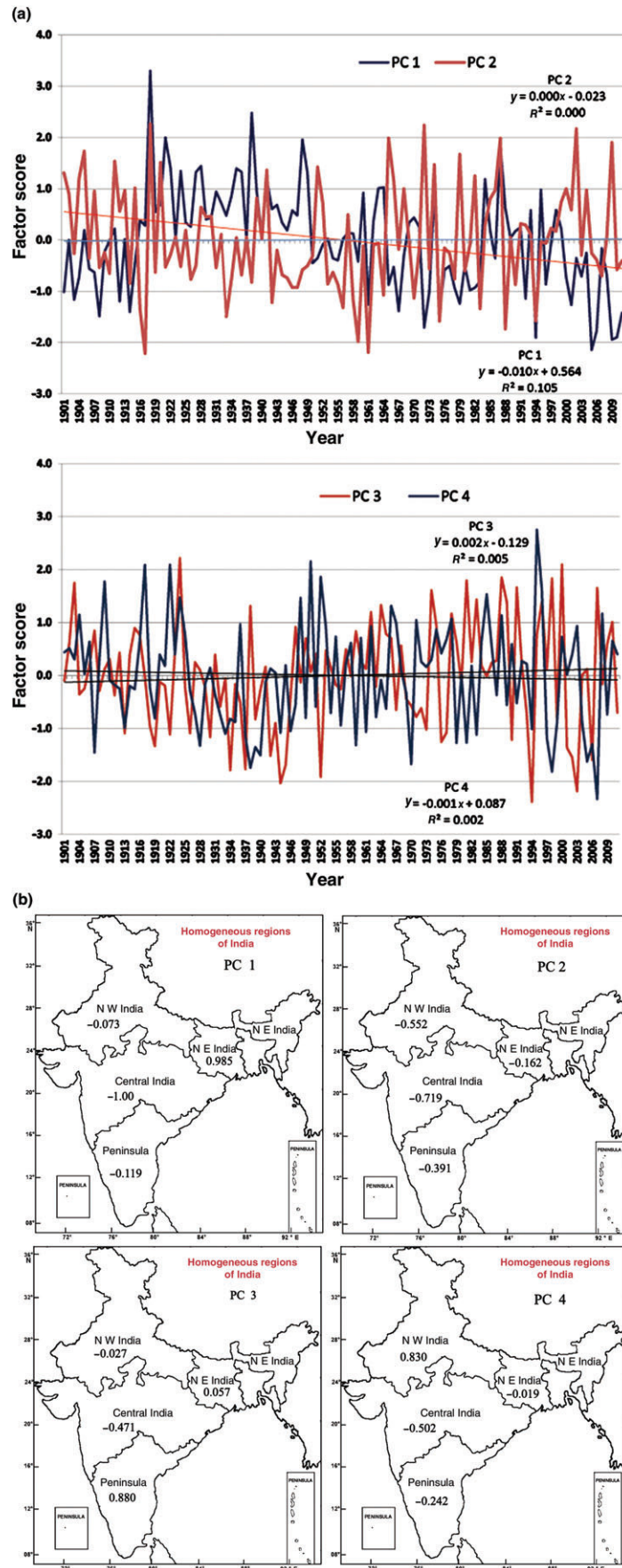


Figure 5. (a) Time series of four principal components. (b) Spatial patterns of four EOFs of all-India southwest monsoon rainfall.

Table 3. Results of the trend analysis of four principal component time series.

	PC 1	PC 2	PC 3	PC 4
1901–1930	Increase**	Decrease*	Decrease*	Decrease
1931–1960	Decrease**	Decrease	Increase*	Increase**
1961–1990	Increase	Increase	Increase	Increase
1991–2011	Decrease**	Increase	Increase	Increase
1901–2011	Decrease**	Increase	Increase	Decrease

**Significant at 99% confidence level. *Significant at 95% confidence level.

homogeneous regions have different phases in the epochal variations. Northeast India (NEI) and Peninsular India have 30-year positive and negative epoch but their phases are opposite. Southwest monsoon rainfall has increased at the rate of $14.8 \text{ mm year}^{-1}$ during 1901–1930 and it decreased almost at similar rate ($15.28 \text{ mm year}^{-1}$) during the period 1991–2011 for Northeast India. However, Northwest India and Central India (CNI) have two consecutive 30-year positive and next two consecutive 30 years as negative epochs. However, their phases are not the same. Southwest monsoon rainfall has shown increasing trend for Northwest India during 1901–1930 and 1931–1960 and decreasing trend during 1961–1990 and 1991–2011. Over Central India, negative trend was observed during 1931–1960 and 1961–1990. The southwest monsoon rainfall of all-India has shown increasing trend in the first two 30-year epoch, i.e. 1901–1930 and 1931–1960 and decreasing trends in 1961–1990 and 1991–2011. The epochal trend of southwest monsoon rainfall is thus clearly noticed in regional scales and in all-India scale. It is also noted that all-India rainfall on these multidecadal 30-year period has fluctuated within $+4.7\%$ (1931–1960 epochal average) to -2.1% (1901–1930 epochal average) of the long-term normal which is not very significant. Therefore, all-India rainfall over the last 111 years has remained quite stable in spite of the minor fluctuations on epochal basis which could be even significant in some epochs as shown in Table 2.

In order to reveal the covariability modes of southwest monsoon rainfall of India and to identify the spatial patterns, we have performed the Empirical Orthogonal Function (EOF) analysis of monsoon rainfall time series of four homogeneous regions referred in Figure 2(a) based

on covariance matrix. The EOF 1, EOF 2, EOF 3 and EOF 4 account 52.6, 32.9, 7.8 and 4.7% of total variance, respectively. The factor coefficients of EOF 1, EOF 2, EOF 3 and EOF 4 are shown in Figure 5(b). The EOF 1 is dominated by NEI with highest coefficient of 0.98. The dipole type spatial structure with positive values in the northeastern parts and negative values over rest of the country noticed here was also noticed earlier by Pant and Rupa Kumar (1997) and Guhathakurta and Rajeevan (2008). In the second EOF, highest magnitude of coefficient is found over CNI. We have performed trend analysis on all the four principal component time series over each of the 30-year epoch, viz. 1901–1930, 1931–1960, 1961–1990 and 1991–2011, and also for the complete period 1901–2011 (Table 3). The results of trend analysis of PC1 are similar to that of rainfall time series of NEI with alternative significant increasing and decreasing trends over the epochs with the exception during the 1961–1990 period. In fact, during the period 1961–1990 neither the principal components (Table 3) nor the four regions rainfall time series (Table 2) showed any significant trend. Significant decreasing trends are noticed for PC2 and PC3 during 1901–1930 and increasing trend for PC4 in 1931–1960. Figure 5(a) shows the time series of all the four PCs for the period 1901–2011.

3.3. Long-term trends

The results of the trend analysis of all-India rainfall for the four monsoon months of the southwest monsoon during the period 1901–1950, 1951–2011 and for the period 1901–2011 are shown in Table 4. The regression equation and the Pearson product moment correlation coefficients are also given.

Rainfall for the individual monsoon months has shown increasing trends during the period 1901–1950. However, increasing trend was not significant for June and August rainfall. But both July and September rainfall showed significant increasing trends (July: 99% level and September: 90% significance level). As a result, southwest monsoon seasonal rainfall during the period 1901–1950 showed a significant (99% level) increasing trend.

During the period 1951–2011, except for the month of June, decreasing trends are observed for all the remaining months and for the season. However, the decreasing trend

Table 4. Slope and Pearson product moment correlation coefficient R squared (RSQ) value for all India and Monsoon Core Region of India (MCRI) rainfall during the period 1901–1950, 1951–2011 and for the period 1901–2011.

Period		Slope					RSQ				
		June	July	August	September	Monsoon	June	July	August	September	Monsoon
All India	1901–1950	0.50	1.33	0.32	0.63	2.79	0.03	0.23	0.02	0.06	0.23
	1951–2011	0.10	–0.49	–0.40	–0.21	–1.00	0.00	0.05	0.04	0.01	0.02
	1901–2011	0.00	–0.07	0.01	–0.03	–0.09	0.00	0.00	0.00	0.00	0.00
MCRI	1901–1950	0.06	1.33	0.20	0.61	2.19	0.00	0.16	0.00	0.04	0.14
	1951–2011	0.07	–0.48	–0.57	–0.54	–1.53	0.00	0.04	0.06	0.05	0.06
	1901–2011	–0.05	–0.21	–0.05	–0.15	–0.45	0.00	0.02	0.00	0.01	0.02

—, decreasing 90% significant; —, increasing 90% significant; —, increasing 99% significant.

was statistically significant (90% level) only for the July rainfall. All-India June rainfall and August rainfall had increased whereas July and September rainfall and the southwest monsoon rainfall had decreased during the whole data period of 1901–2011. These results are consistent with the studies on changes in the global monsoon rainfall by Zhou *et al.* (2008a) revealing decreasing trends in South Asian monsoon (considering June, July and August rainfall). Zhou *et al.* (2008b) have suggested this weakening is due to tropical ocean warming.

Table 4 also depicts the results of trend analysis of rainfall of the MCRI for four monsoon months and for the southwest monsoon season during the period 1901–1950, 1951–2011 and for the period 1901–2011. July rainfall and southwest monsoon rainfall of this region had increased significantly (95%) during the period 1901–1950. August rainfall and southwest monsoon rainfall had decreased significantly (at 90% level) during the period 1951–2011. Other studies by Zhang and Zhou (2011) and Zhou *et al.* (2008a) have also reported decreasing trends in Asian monsoon during this period. However during the complete period of 1901–2011, as in the case of all-India rainfall, no significant increase or decrease has been noticed in monthly or seasonal scale.

4. Changes in the frequency of days with rainfall of different intensities

Frequency of days with rainfall of different intensities, viz. (1) wet day, (2) rainy day and (3) heavy rainfall day are computed from the daily rainfall data of all available stations for four monsoon months and monsoon season. The trend analysis is then performed on the time series of frequencies considering three different periods, viz. 1901–1950, 1951–2011 and the complete period 1901–2011. As there are many missing years and the availability of the data for all the stations are not same in each period, we have selected those stations which have at least 50% or more available years of data in each period. To know the distribution of properties of trend spatially or more specifically for each of the 36 meteorological subdivisions, we have calculated the number of stations having decreasing/increasing trends in percentage of total stations considered for each subdivision. The consideration of frequency of station in percentage is due to unequal number of stations in three periods (7598 stations in period 1951–2011, 2672 stations in the period 1901–1950 and 7650 stations in the period 1901–2011).

Table 5 shows the percentage frequency of number of stations in each subdivision having increasing (+ve) trend and decreasing (–ve) trend in wet days/rainy days/heavy rainfall days during southwest monsoon. The figures in bold in ninth column indicate that the percentage of stations with increasing trend in heavy rainfall in these subdivisions are more than the percentage of stations with decreasing trend.

In order to see the spatial pattern of the changes in the frequency of rainfall events, the subdivisions are shaded by different colours to indicate more positive and negative trends. Figure 6 shows the spatial pattern of changes in rainfall events for the period 1901–1950 and 1951–2011. Figure 7 shows the same as that of Figure 6 but for the complete period. For frequency of wet day, 17 of 36 subdivisions had more number of positive trends than negative trends during the period 1901–1950 (Figure 6(a)) whereas 6 subdivisions had more number of positive trends than negative trends during the period 1951–2011 (Figure 6(b)). For frequency of rainy days, 17 of 36 subdivisions had more number of positive trends than negative trends during the period 1901–1950 (Figure 6(c)) whereas 3 subdivisions had more number of positive trends than negative trends during the period 1951–2011 (Figure 6(d)). For frequency of heavy rainfall days, 18 of 36 subdivisions had more number of positive trends than negative trends during the period 1901–1950 (Figure 6(e)) whereas 11 subdivisions had more number of positive trends than negative trends during the period 1951–2011 (Figure 6(f)). Thus in the period 1901–1950 increasing trends in the frequencies of all the categories of rain were more than the period 1951–2011.

In the recent period (1951–2011), positive (increasing) trends in heavy rainfall frequency was more for the subdivisions: Jharkhand, Orissa, Gangetic and Sub Himalayan West Bengal, Arunachal Pradesh, Gujarat, Saurashtra & Kutch, Marathwada, Telangana, Rayalaseema and West Rajasthan, whereas the frequency of decreasing trends in heavy rainfall are more in most part of Central India. In other words, we can say the intensity of extreme rainfall events have decreased significantly over the central region of India during the period 1951–2011 whereas increased over Eastern India, parts of South Peninsula and extreme western parts of the country.

Consideration of the complete period 1901–2011 (Figure 7(c)) also shows that the intensity of extreme rainfall events (i.e. heavy rainfall events here) have decreased significantly in the central region whereas it has been increased in Peninsular India, coastal regions of Eastern India and also Eastern Rajasthan, Punjab, Haryana, Maharashtra, Gujarat and Nagaland, Manipur, Mizoram and Tripura. Rainfall of small intensities (wet day and rainy day) show more decreasing trends in most of the areas/subdivisions of the country except Madhya Maharashtra, north and south interior Karnataka. The desert regions of India, i.e. West Rajasthan, shows increasing trend in rainfall of very small intensities. Earlier studied by Guhathakurta *et al.* (2011b) using station data also obtained increasing trends in the frequency of heavy rainfall events over western coast and parts of Peninsular India and decreasing trends in rainy days over most parts of the country.

As we have already shown, there exists large discrepancies between the IMD datasets based on station data and gridded datasets both in climatology and in time evolution. These changes may reflect more in the study of frequency of heavy rainfall events than the total rainfall

Table 5. Percentage frequency of number of stations in each subdivision having increasing (+ve) trend and decreasing (–ve) in wet days/rainy days/heavy rainfall days during southwest monsoon.

Subdivision No.	No. of stations	Percentage of stations during 1901–1950			Percentage of stations during 1951–2011			Percentage of stations during 1901–2011	
		+ Trend	– Trend	No. of stations	+ Trend	– Trend	No. of stations	+ Trend	– Trend
1	1	0/0/0	0/0/0	7	0/0/0	85.7/85.7/0	7	0/0/0	14.3/14.3/0
2	3	0/0/0	0/0/0	50	0/0/2	0/0/0	51	0/0/0	0/0/0
3	105	7.6/1/6.7	34.3/21.9/7.6	217	0.9/1.4/1.9	6/5.1/1.9	219	0.5/0/2.8	30.1/25.6/6.5
4	28	7.1/7.1/14.3	14.3/7.1/0	96	0/1/1.1	6.3/3.1/2.2	96	0/1/2.2	10.4/7.3/1.1
5	27	11.1/0/3.7	29.6/11.1/3.7	60	3.3/3.3/6.7	8.3/6.7/1.7	61	1.6/0/1.6	24.6/18/6.6
6	116	6/4.3/3.4	19.8/15.5/6	154	2.6/0.6/1.9	3.9/3.2/0.6	166	0.6/0.6/2.4	20.5/19.3/1.8
7	119	5.9/0.8/7.6	35.3/24.4/0.8	377	12.7/15.4/11.5	3.2/2.4/0.8	378	0.5/0.3/4	17.2/9.3/1.1
8	97	19.6/13.4/6.2	9.3/8.2/2.1	209	2.4/3.3/8.2	16.7/9.6/6.7	210	0/0/1.9	24.8/19/9.1
9	147	7.5/5.4/5.4	21.8/17/8.8	428	3.7/2.8/3.7	11.9/8.9/3.7	428	0.2/0/2.1	19.2/16.6/4.2
10	126	32.5/19/5.6	4/3.2/1.6	195	1/0.5/2.1	47.2/35.9/18.8	195	0.5/0/1	60/57.4/22
11	127	22/18.9/7.1	3.1/0/3.9	172	1.2/0.6/2.4	48.8/39.5/27.8	172	0/0/2.4	71.5/54.1/16.6
12	39	20.5/12.8/2.6	2.6/0/7.7	98	0/0/1.1	42.9/42.9/24.7	99	0/0/0	35.4/34.3/24.5
13	52	7.7/1.9/0	15.4/9.6/3.8	103	6.8/1.9/3	3.9/5.8/5	104	3.8/1.9/8.9	9.6/4.8/1
14	40	15/10/5	5/2.5/0	94	0/0/0	22.3/22.3/14	94	0/0/3.2	19.1/14.9/0
15	19	0/5.3/0	10.5/0/0	106	2.8/0.9/2	30.2/30.2/18.4	106	0.9/0/0	17/17/7.1
16	41	7.3/0/0	24.4/19.5/2.9	72	4.2/1.4/0	4.2/1.4/0	83	1.2/1.2/0	9.6/9.6/0
17	66	1.5/0/3.1	16.7/6.1/0	149	5.4/2/2.2	4/4.1/0	159	8.2/2.5/1.4	4.4/3.2/1.4
18	136	10.3/7.4/2.2	2.2/0.7/0.7	375	10.7/3.7/1.9	4.8/4/1.9	378	7.1/1.9/2.7	9/3.2/0.8
19	110	32.7/39.1/22.7	1.8/0.9/1.8	212	1.4/0.9/1.4	27.4/19.3/8.1	213	5.2/3.3/1	22.1/10.8/3.3
20	88	25/20.5/8	0/0/0	159	0/0/1.9	35.8/27/4.4	159	0/0/1.3	44.7/31.4/8.2
21	83	4.8/0/6	2.4/0/0	238	0/0.4/2.1	25.6/3.4/0.8	238	0/0/5.1	26.1/3.4/0
22	30	3.3/0/10	13.3/0/0	165	1.8/0.6/1.2	13.9/3/0.6	165	0.6/0.6/1.2	9.1/0.6/0
23	51	15.7/2/11.8	2/0/0	76	0/5.3/0	18.4/5.3/26.3	77	2.6/0/16.9	24.7/3.9/5.2
24	110	9.1/14.5/11.8	9.1/0/0	166	1.8/2.4/1.2	27.1/7.2/4.2	167	1.8/8.4/4.8	39.5/4.2/1.2
25	8	12.5/0/0	0/0/0	70	1.4/0/2.9	12.9/11.4/0	70	1.4/0/4.3	4.3/2.9/0
26	88	35.2/28.2/34	0/0/0	160	1.3/0.6/0	18.1/11.9/3.1	160	0/0/1.3	35.6/12.5/6.3
27	74	23/14.9/6.8	2.7/1.4/0	117	0/0/1.7	46.2/48.7/20.7	117	0.9/0/0	46.2/47/12.8
28	103	2.9/3.9/1	33/8.7/1.9	490	3.5/1.4/1.5	7.3/6.1/1.7	490	1.4/1/1	7.6/2.4/0.6
29	64	3.1/3.1/0	3.1/0/0	460	0.2/0.2/1.3	15.7/10.2/0.9	462	0.2/0/1.8	9.3/5.2/0
30	71	0/0/1.4	23.9/2.8/0	291	4.8/4.5/0.4	7.2/2.4/0	291	1.4/1.7/0.4	11.7/2.4/0
31	235	5.1/0.9/1.3	22.1/10.2/1.3	612	1.6/1.5/1.3	23/25.2/2.9	612	1.5/1.3/2.2	29.4/24.2/2.4
32	22	36.4/13.6/0	0/0/0	139	1.4/0/1.4	7.2/1.4/2.9	139	0/0/0.7	7.2/0/0
33	49	16.3/6.1/0	18.4/4.1/0	440	7.7/2/0.9	3.9/2.5/1.2	440	4.1/1.8/0.5	3.4/0.5/0
34	115	1.7/3.5/0	27.8/6.1/0.9	725	23.3/16.7/2.3	1.9/0.4/3.7	725	5.9/2.1/1.3	3.2/1.7/0.6
35	80	16.3/3.8/1.3	5/3.8/5	111	0/0/0	31.5/25.2/17.1	114	0.9/0.9/0.9	59.6/49.1/25.4
36	2	50/50/50	0/0/0	5	0/0/0	0/0/0	5	20/20/0	0/0/0

The figures in bold in ninth column indicate that the percentage of stations with increasing trend in heavy rainfall in these subdivisions are more than the percentage of stations with decreasing trend.

(Figure 3). Goswami *et al.* (2006) considering IMD 1×1 DEG GRID DATA for the period 1951–2000 have found significant decreasing trend in the frequency of moderate rainfall events ($5 \text{ mm} \leq \text{daily rainfall} < 100 \text{ mm}$) and significant increasing trends in the frequency of heavy rainfall events ($\text{daily rainfall} \geq 100 \text{ mm}$) and very heavy rainfall events ($\text{daily rainfall} \geq 150 \text{ mm}$) over the region within $74.5^\circ\text{--}86.5^\circ\text{E}$ and $16.5^\circ\text{--}26.5^\circ\text{N}$ (Figure 2(b)). Using the 1° latitude $\times$ 1° longitude gridded data, Rajeevan *et al.* (2008) have also shown significant increasing trends in very heavy rainfall events over the same region during the period 1951–2004 but the rate of increase is slightly lesser than that obtained by Goswami *et al.* (2006) during the period 1951–2000. In order to compare our findings based on station data with these results based on

gridded data, we have considered the same data period (1951–2000) and also computed total number of events for moderate rain ($5 \text{ mm} \leq \text{daily rainfall} < 100 \text{ mm}$), heavy rainfall ($\text{daily rainfall} \geq 100 \text{ mm}$) and very heavy rainfall ($\text{daily rainfall} \geq 150 \text{ mm}$) over these region in each year of southwest monsoon season (June–September) by considering all the stations over this region. Figure 8 shows the temporal variation in the number of moderate wet day ($5 \text{ mm} \leq \text{daily rainfall} < 100 \text{ mm}$), heavy rainfall events ($\text{daily rainfall} \geq 100 \text{ mm}$) and very heavy rainfall events ($\text{daily rainfall} \geq 150 \text{ mm}$) and their trend lines. Two separate sets of trend lines are drawn to show the behaviour of trends during the period 1951–2000 in black colour and during the period 1951–2010 in red colour. During the period 1951–2000, frequency of moderate rain events

TREND IN SOUTHWEST MONSOON RAINFALL OVER INDIA

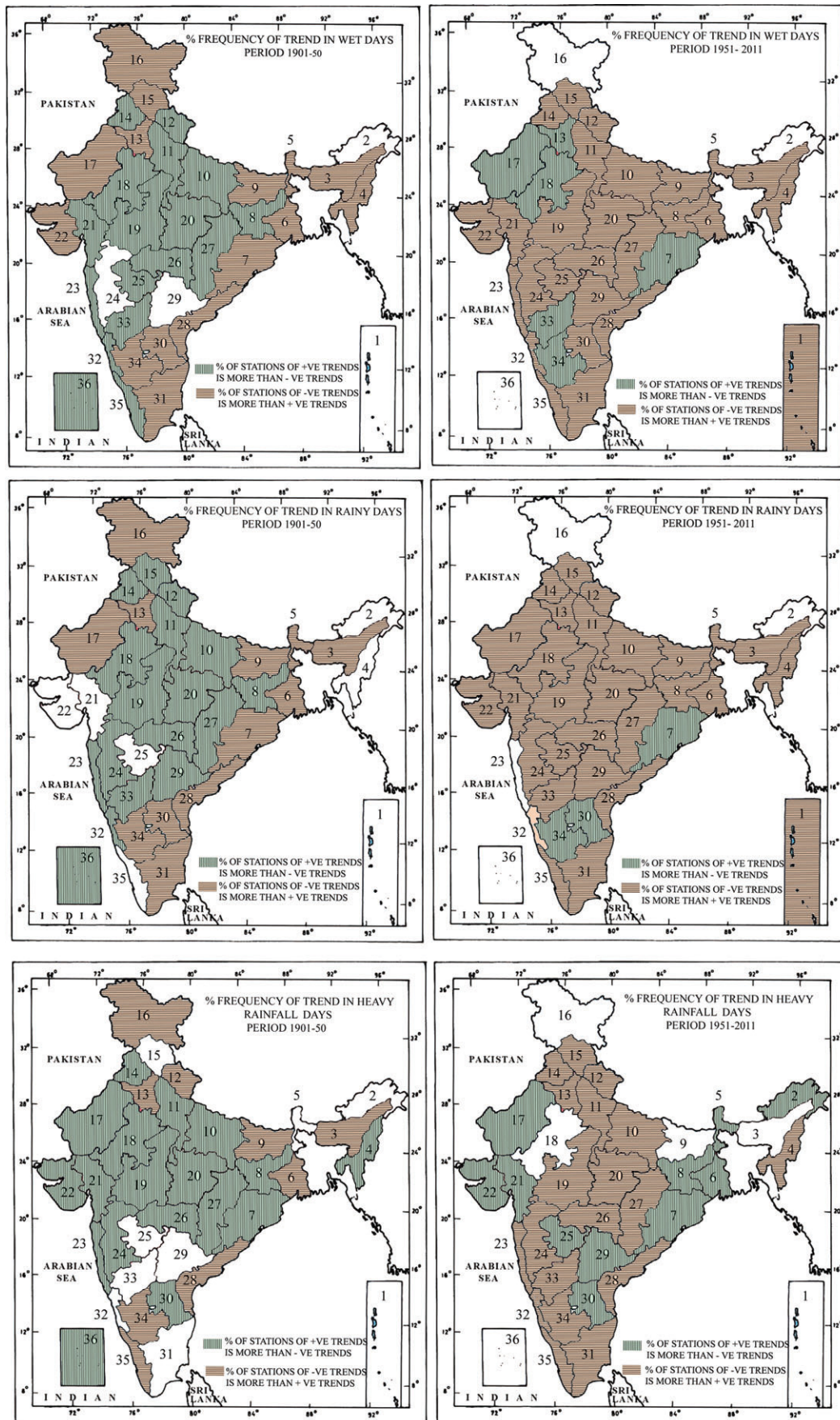


Figure 6. Subdivisions showing the frequency of stations having positive (increasing) trends more or less than negative (decreasing) trends during the period 1901–1950 for (a) wet days, (c) rainy days and (e) heavy rainfall days and 1951–2011 for (b) wet days, (d) rainy days, and (f) heavy rainfall days.

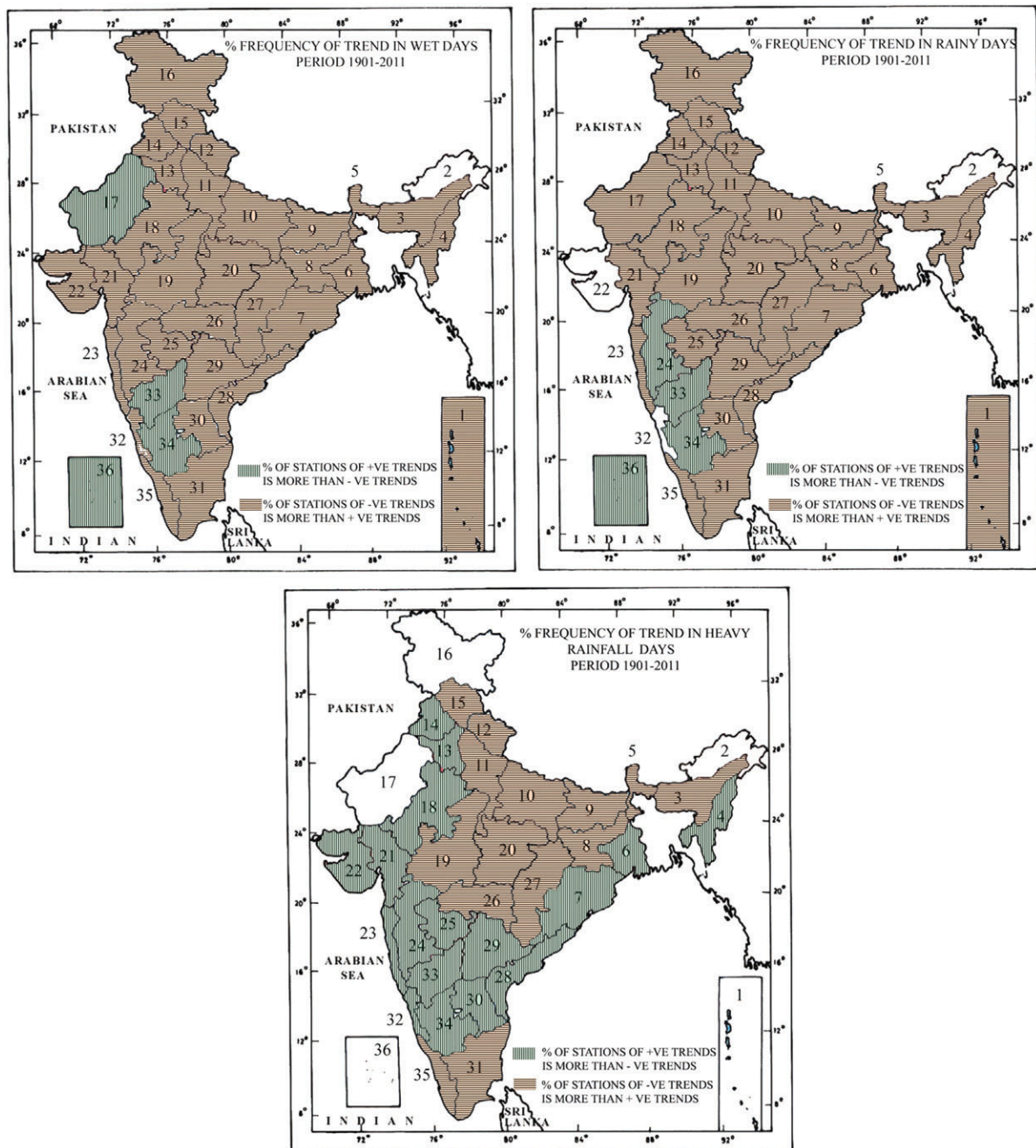


Figure 7. Spatial distribution of the subdivisions showing the frequency of stations having positive (increasing) trends more or less than negative (decreasing) trends during the period 1901–2011 for (a) wet days, (b) rainy days and (c) heavy rainfall days. Vertical lines shows more positive trends and horizontal lines shows more negative trends.

and heavy rain events (daily rainfall ≥ 100 mm) have not shown any significant trend (not even 90% confidence level) whereas frequency of very heavy rainfall events (daily rainfall ≥ 150 mm) has shown significant increasing trends (at 99% confidence level). These results are little different from that of Goswami *et al.* (2006) where they have observed using gridded data, significant decreasing trends in moderate rain events and significant increasing trends in heavy and very heavy rain events. But both station data analysis and gridded data analysis suggested increasing trends in very heavy rainfall events during the period 1951–2000. But inclusion of recent data, i.e. up to 2010,

indicates no significant trends in heavy and very heavy rainfall events. But frequency of moderate rainfall events has significant decreasing trends during 1951–2010.

5. Change-point detection in southwest monsoon rainfall

There are earlier scientific results of shift in global climate around 1970s (Trenberth and Hurrell, 1994; Zhou *et al.* 2008a). Also as mentioned earlier there were several findings (Zhang and Zhou, 2011, etc.) regarding weakening of Indian monsoon around 1950s. It is therefore necessary to

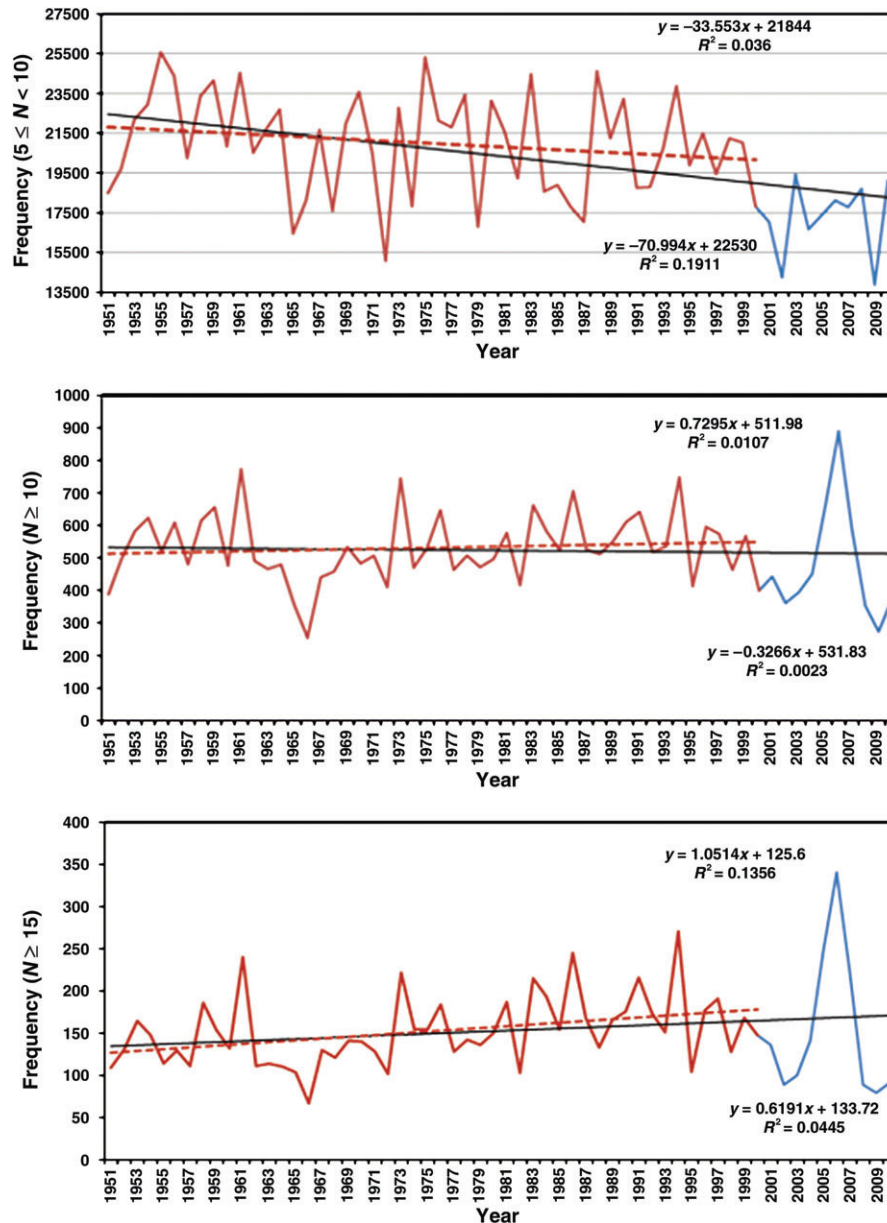


Figure 8. Temporal variation in the total number of events for moderate rain ($5 \text{ mm} \leq \text{daily rainfall} < 100 \text{ mm}$), heavy rainfall (daily rainfall $\geq 100 \text{ mm}$) and very heavy rainfall (daily rainfall $\geq 150 \text{ mm}$) and their trend lines for the periods 1951–2000 and 1951–2010 over the Monsoon Core Region of India.

know the change points in southwest monsoon of smaller regional scale. The Pettitt test was used to detect the change point in southwest monsoon rainfall over India. Figures 9–13 show the result of Pettitt test in identifying the change point for four homogeneous regions, *viz.* Northwest India, Central India, Peninsular India, Northeast India, and all-India, respectively. Significant upward shift in southwest monsoon rainfall is detected for Northwest India in the year 1942 and Peninsular India in the year 1946, whereas Central India had also experienced upward shift but not significant in the year 1926. Northeast India got significant downward shift in southwest monsoon rainfall in the year 1961 and as a result all-India southwest monsoon rainfall had experienced significant downward shift in the year 1965.

6. Conclusions

The rainfall data of longer series (1901–2011) was analysed to study the temporal and spatial variability of rainfall pattern in India. The multidecadal variability observed here is closely linked with the variability of other global or regional parameters observed by several studies. The decadal and inter decadal changes and variability of Indian monsoon rainfall are well connected with Indian Ocean sea surface temperature (SST), El Niño–Southern Oscillation (ENSO), etc. (Torrence and Webster, 1999; Ashok *et al.*, 2004, etc.). Possibility of decadal scale variability in Indian monsoon rainfall influenced by equatorial Pacific Ocean SST was suggested by Kripalani and Kulkarni (2001) and Krishnamurthy and Goswami (2000). It has also been reported that the impact of El Niño (La Niña)

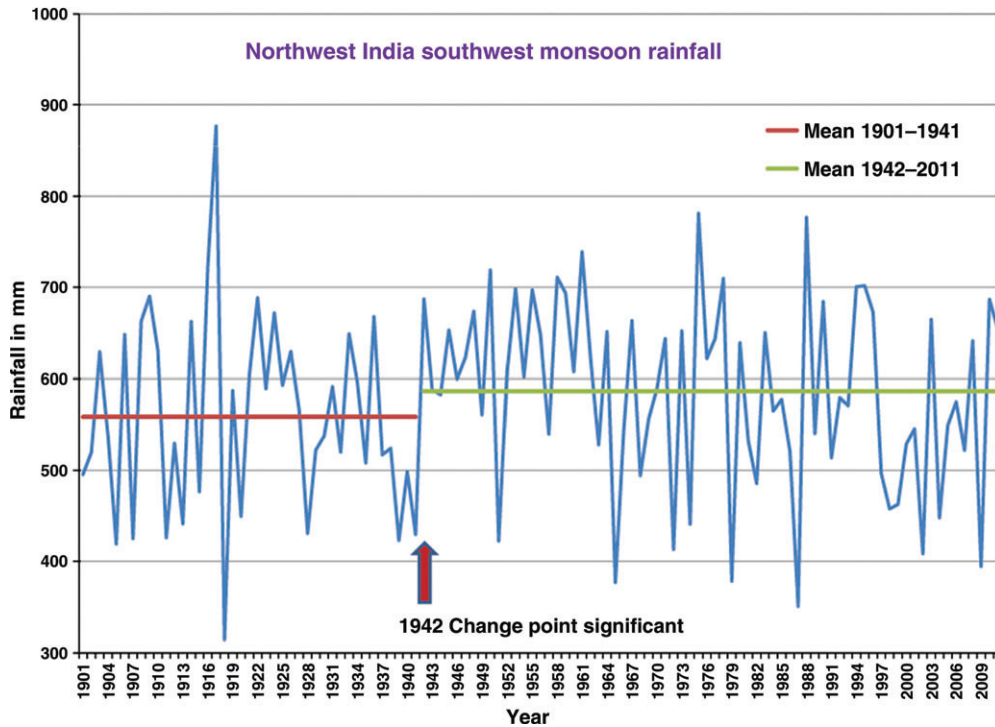


Figure 9. Climate shift or change point in monsoon rainfall over Northwest India.

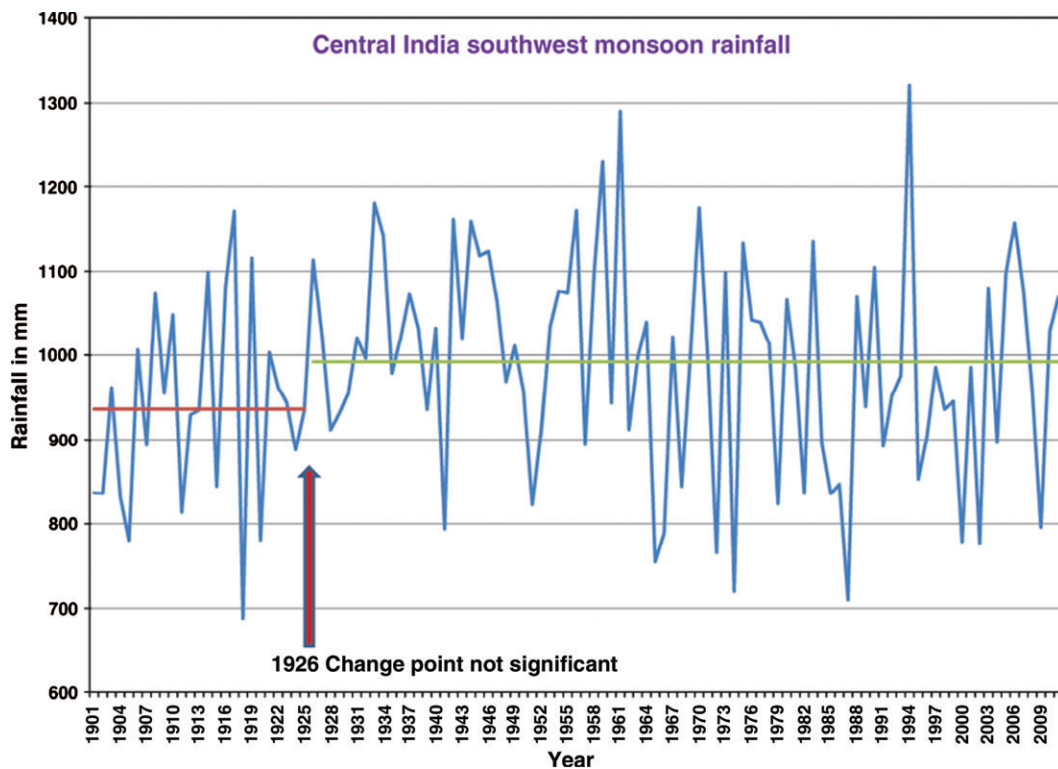


Figure 10. Climate shift or change point in monsoon rainfall over Central India.

is more severe during the below-normal (above-normal) rainfall epochs. Krishnamurthy and Krishnamurthy (2013) have shown that the decadal variability of Indian monsoon rainfall is associated with the decadal variability of the North Atlantic and North Pacific Oceans. The weakening of ENSO–Indian monsoon relationship has been observed

during the last 30 years. The physical mechanisms governing the decadal changes of the Indian monsoon circulation has been better described by Kucharski *et al.* (2006). The Indian monsoon rainfall has the strongest relationship with the equatorial eastern Pacific SSTs (Webster *et al.*, 1998), but Indian Ocean SSTs appear also to play a role

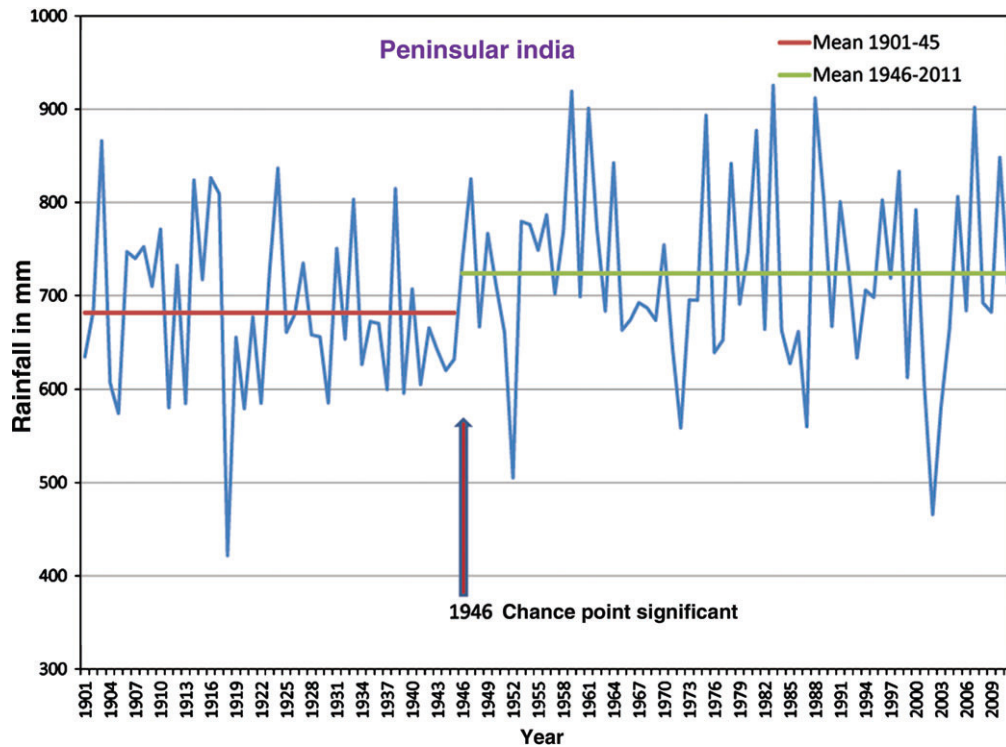


Figure 11. Climate shift or change point in monsoon rainfall over Peninsular India.

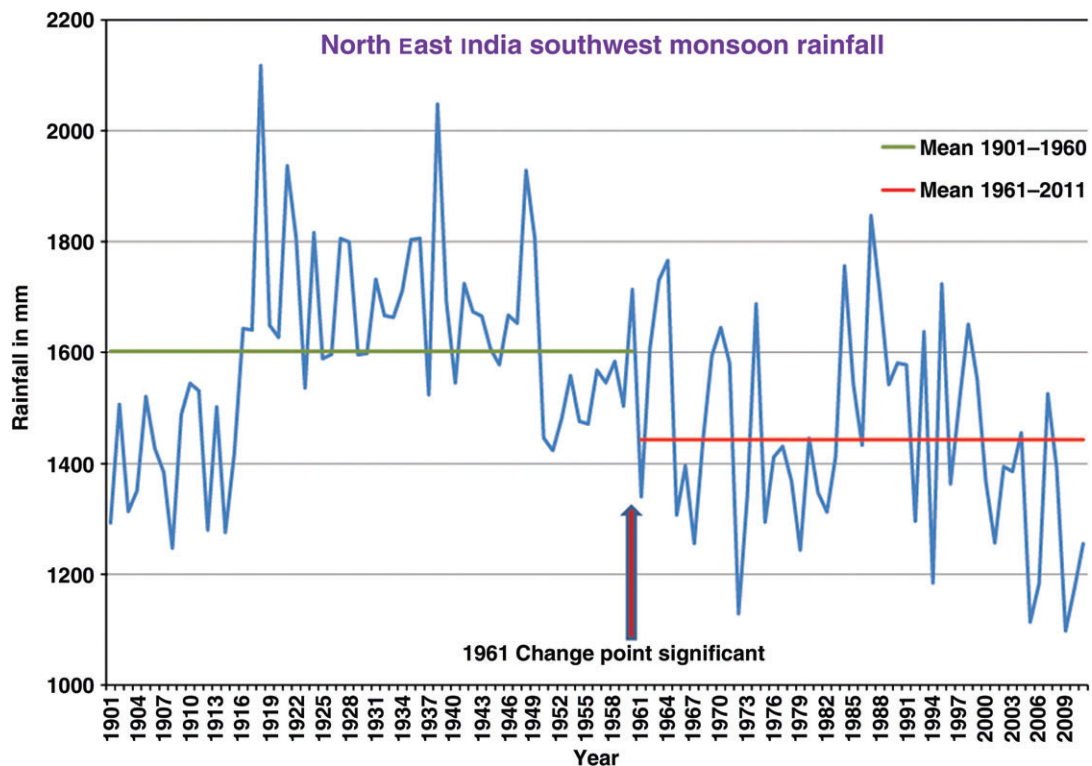


Figure 12. Climate shift or change point in monsoon rainfall over Northeast India.

(Terry *et al.*, 2005; Clark *et al.*, 2000; Li *et al.*, 2001; Ashok *et al.*, 2004; Lau and Wu, 2001) in influencing Indian monsoon. Kucharski *et al.* (2006) have shown that using ensemble integrations of an Atmospheric General Circulation Model (AGCM) forced globally with observed

SST anomalies, the observed decadal variability of Indian monsoon rainfall can be reproduced, although amplitudes are somewhat overestimated. Using a classification for ENSO and Indian Ocean Dipole (IOD) events for the past 130 years (1877–2006), Ummenhofer *et al.* (2011)

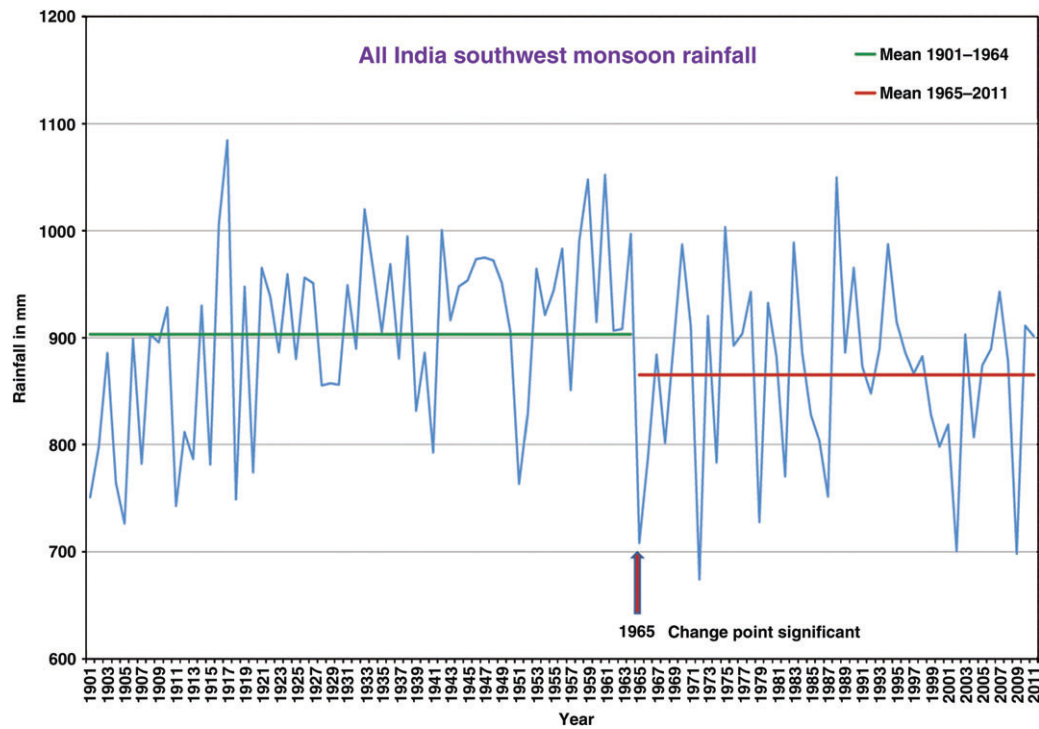


Figure 13. Climate shift or change point in all-India monsoon rainfall.

Table 6. Decadal monsoon rainfall (percentage departure from mean) of all-India and four homogeneous regions, number of positive IOD (PIOD) years and EL NINO years and their correlations.

Decades	No. of PIOD years	No. of El Nino years	All-India rainfall	Northwest India rainfall	Central India rainfall	Northeast India rainfall	Peninsular India rainfall
1901–1910	2	3	–6.1	–1.8	–5.9	–7.9	0.2
1911–1920	2	3	–2.9	–4.8	–3.4	2.6	–4.8
1921–1930	1	3	2.6	1.2	–1.3	11.7	–3.8
1931–1940	1	1	4.7	–4.6	6.3	12.4	–2.5
1941–1950	3	0	5.8	6.2	5.9	9.5	–2.8
1951–1960	1	2	3.8	8.1	4.7	0.2	3.9
1961–1970	3	3	0.6	0.1	0.3	–1.3	3.9
1971–80	2	2	–2.0	2.8	–1.0	–8.9	–0.2
1981–1990	2	3	–0.7	–1.4	–4.5	1.2	4.2
1991–2000	3	2	–1.1	–1.4	–2.6	–2.8	3.6
2001–2010	2	3	–5.0	–5.7	0.6	–15.1	–1.9
Correlation with no. of PIOD years			–0.2	0.0	–0.2	–0.3	0.3
Correlation with no. of El Nino years			–0.7	–0.4	–0.8	–0.5	0.2

have expanded previous observational studies and have shown that decadal variations in Indian Ocean conditions modulate the ENSO–Indian monsoon relationship. In our present analysis, we have shown the existence of multi-decadal variability of four homogeneous regions of India and their phases are different. The multidecadal variability of all-India rainfall is the results of these four regions variability. Strength and phases of multidecadal variability of four homogeneous regions of India modulate the phase of all-India rainfall. Decadal rainfall of all-India and four homogeneous regions along with number of positive IOD years and EL NINO years in each decades are shown in Table 6. Except for Peninsular India all other regions have negative correlation with number of positive IOD years. Similarly only rainfall of Peninsular India

has positive correlation with EL NINO years whereas other three homogeneous regions as well as all-India rainfall have strong negative correlations. Further analysis is required to link the multidecadal variability of global or regional parameters with multidecadal variability of monsoon rainfall of these homogeneous regions.

In this article, we have shown that there was significant increasing trend in all-India southwest monsoon rainfall as well as in July and September months for the period 1901–1950. However, during the recent period, i.e. 1951–2011, all-India monsoon rainfall as well as July, August and September rainfall have shown decreasing trends but the trend was significant only for July rainfall. During the complete period of 111 years, all-India rainfall did not show any significant trend for the monsoon months

as well as for the whole season. However, rainfall averaged for July and August has shown significant decreasing trend (99% significant level) similar to the result obtained in (Bawiskar, 2009).

The rainfall analysis has been performed separately for MCRI consisting of 230 districts (smaller administrative region) as shown in Figure 2(b). This region is similar to the region chosen by Goswami *et al.* (2006). Rainfall in this region has shown significant increasing trend for the July month and for the monsoon season during the period 1901–1950 but significant decreasing trend in August and monsoon season during 1951–2011. In fact except June, all other months showed decreasing trend in rainfall over this region during the period 1951–2011. The decadal variability of all-India southwest monsoon rainfall showed negative departure during the first two decades followed by positive departure in five decades (1921–1930, 1931–1940, 1941–1950, 1951–1960 and 1961–1970). The recent four decades were in the negative side and a polynomial fitting indicates that the current decade (1911–1920) will also be in the negative side with mean value of -3.8% .

Epochal character of all-India rainfall as well as four homogeneous region southwest monsoon rainfall is clearly noted. The phases of positive and negative trends are different for four regions, *viz.* Northwest India, Central India, Northeast India and Peninsular India. Over Northwest India, positive trends are observed in the first two 30-year epoch, *i.e.* 1901–1930 and 1931–1960, with significant trend in 1931–1960. In the next two epochs, *i.e.* 1961–1990 and 1991–2011, trends are negative but not significant. In Central India, positive trend was observed during 1901–1930 followed by negative trend in 1931–1960 and 1961–1990 and again positive trend in 1991–2011. For Northeast India, significant positive trend was observed in 1901–1930, significant negative trend in 1931–1960 followed by positive trend in 1961–1990 and again significant negative trend in 1991–2011. Monsoon rainfall in Peninsular India has shown negative trend in 1901–1930, positive trend in 1931–1960, negative trend in 1961–1990 and finally positive trend in 1991–2011. But none of the trends over peninsular India are significant at 95% level. As a result, all-India southwest monsoon rainfall has positive trend in the first two epochs and negative trend in the recent two epochs with trend (positive) being significant for one 30-year epoch during 1901–1930 and for other 50-year epoch in 1901–1950. For the whole period (1901–2011), all-India southwest monsoon rainfall trend is not significant at 95% level. Only for Northeast India, the long-term rainfall trend for the period 1901–2011 is significant at 95% level.

Analysis of frequency of rainfall of occurrences of different categories shows that frequencies of wet day and rainy day (rainfall of small amount in a day) have decreased most parts of the country except some regions of west peninsular India, *viz.* Madhya Maharashtra, north and south interior Karnataka and Lakshadweep. In the large regions of the country, *i.e.* 17 of 36 subdivisions, *viz.* Coastal Karnataka, north and south interior Karnataka,

Gujarat region, Saurashtra & Kutch, Konkan & Goa, Madhya Maharashtra, Marathwada, Rayalaseema, Telangana, Coastal Andhra Pradesh, Orissa, Gangetic West Bengal, Nagaland, Manipur, Mizoram & Tripura, East Rajasthan, Haryana, Delhi & Chandigarh and Punjab, increasing trends in the frequency of heavy rainfall events dominated during the period 1901–2011. However in the recent period (1951–2011), only in the 17 subdivisions, increasing trends in the frequency of heavy rainfall were noted.

Over the MCRI significant increasing trends in the frequency of very heavy rainfall events (daily rainfall ≥ 150 mm) are noticed during the period 1951–2000. No significant trends in the frequency of moderate rainfall events ($5 \text{ mm} \leq \text{daily rainfall} < 100 \text{ mm}$) or heavy rainfall events (daily rainfall ≥ 100 mm) are noticed. However, the inclusion of recent data (2001–2010), *i.e.* data for the period 1951–2010, reveals losing of the significant level in the increasing trends of very heavy rainfall events. In fact no significant trends in the frequency of heavy and very heavy rainfall events are noticed during the period 1951–2010 over MCRI. But the frequency of moderate rainfall events ($5 \text{ mm} \leq \text{daily rainfall} < 100 \text{ mm}$) has decreased significantly during the period 1951–2010.

EOF analysis reveals the spatial variability of southwest monsoon rainfall. The first EOF accounting 52.6% of variability is highly dominated by northeast region. The results of trend analysis of EOF trend series is similar to that of Northeast India rainfall time series giving significant decreasing trends. The second EOF accounting 32.9% of variability is mostly influenced by Central India. The Pettitt test detected significant upward shift in southwest monsoon rainfall for Northwest India and Peninsular India in mid 1940s and significant downward shift for Northeast India. Northeast India dominated the all-India rainfall and significant downward shift is noticed in all-India southwest monsoon rainfall in mid 1960s.

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